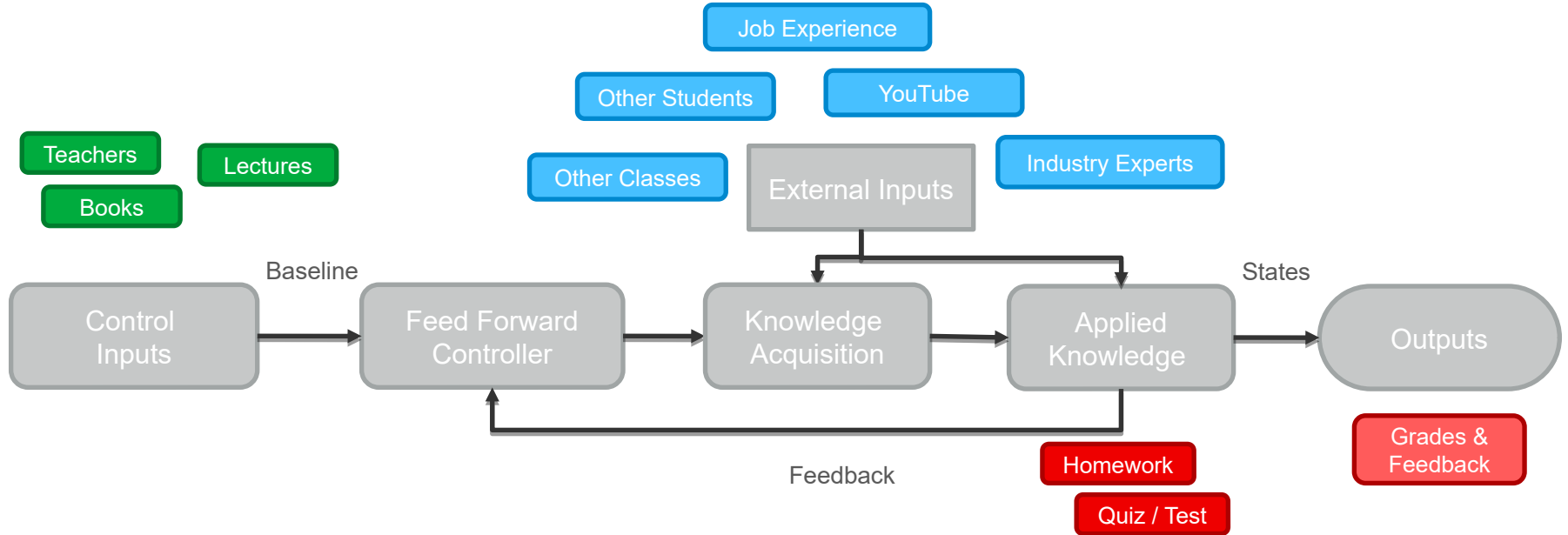
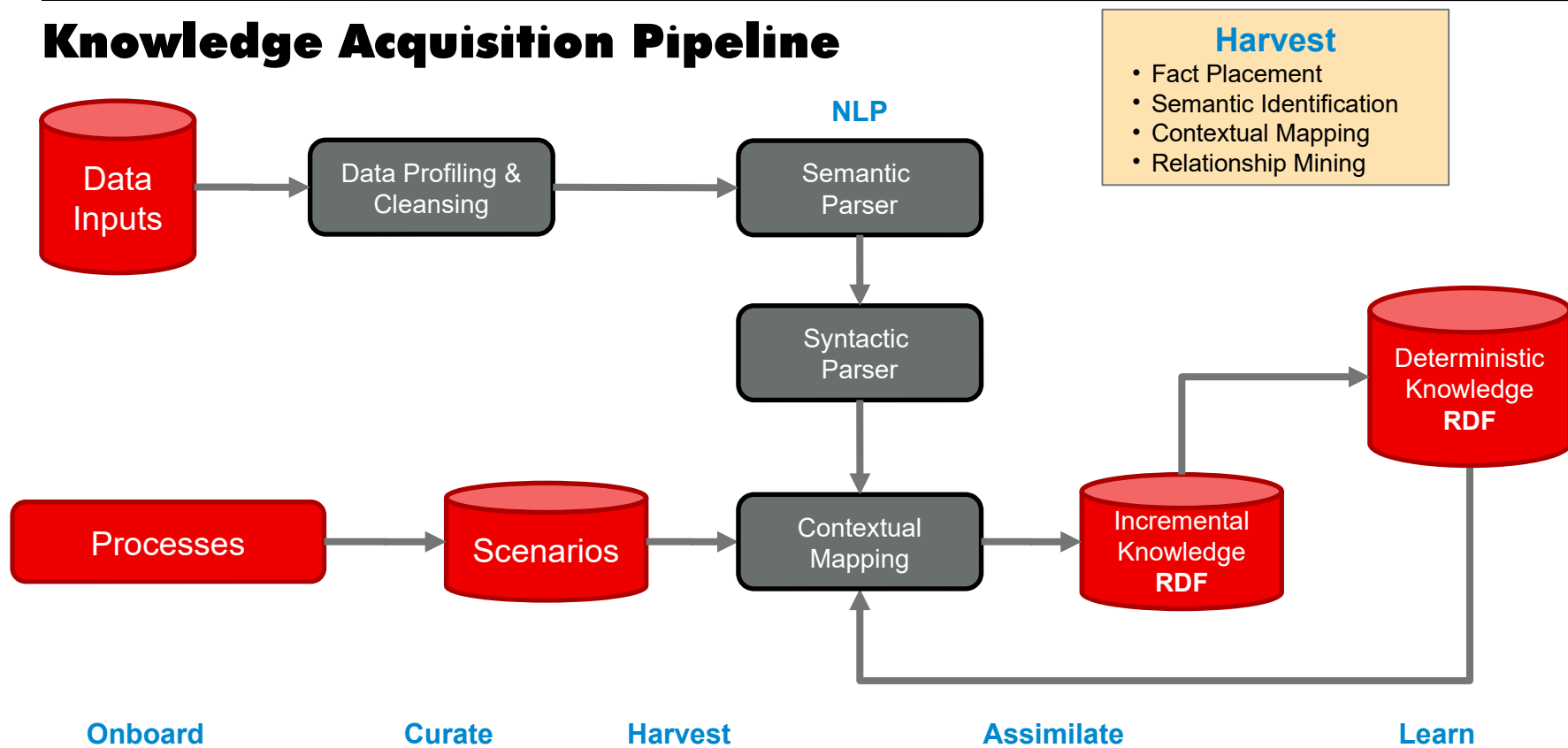


Digital Twin for a Student (Discrete Twin for Education)

(Control Theory + Theory of Constraints + Complexity Theory + Game Theory)



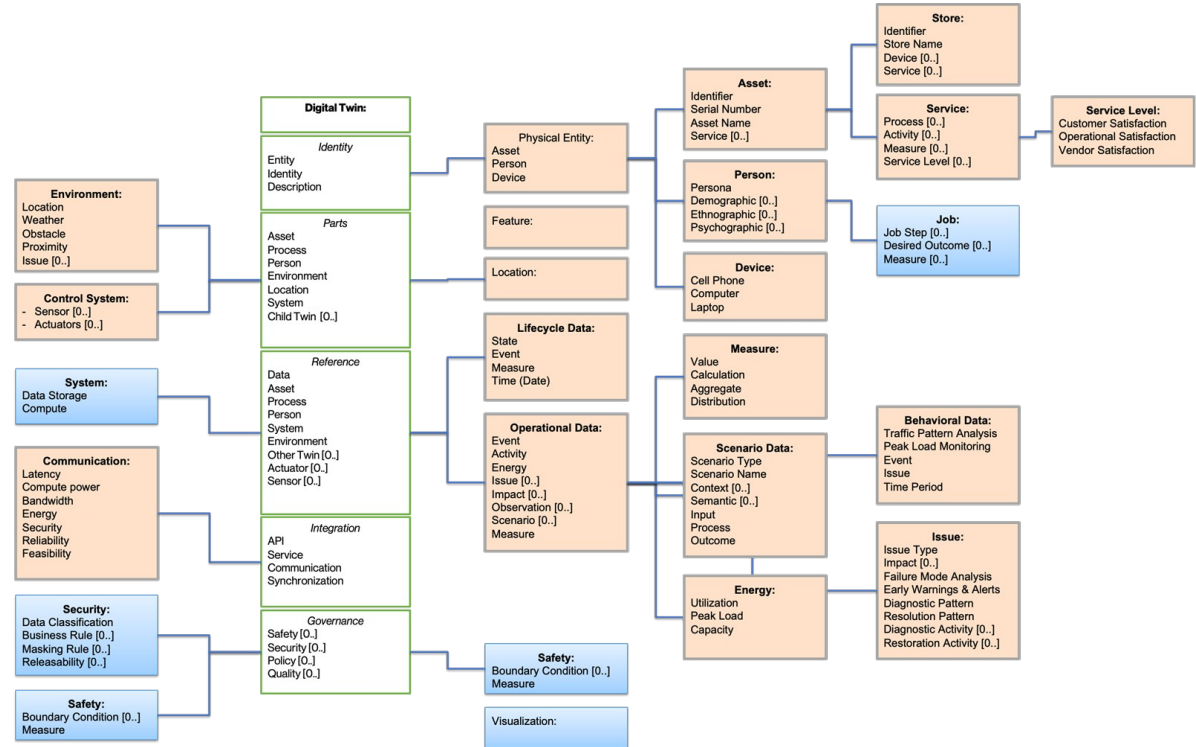
Knowledge Acquisition Pipeline



Creating an upper ontology for a DT Reference Model

Class models can be setup
in the upper ontology in
Metaphactory

- Ontologies can be built as for any type of Twin.
- A proposed reference model (upper ontology) for any type of DT can be instantiated in an RDF
- Specific instances of each Twin can be built in a single instance of an RDF or maintained in separate RDFs to optimize performance.
- Interoperability is enabled because we have adhered to the SKOS model and aligned instances to our ontology scaffold



Assimilation

Assimilating incremental knowledge into Deterministic Knowledge (ontologies)

- an entirely new concept
- an existing concept with a new relation + scenario(s)
- an existing concept with a new attribute + scenario(s)

Insertion into Deterministic Knowledge (ontologies) requires evaluating

- Equality
- Equivalence
- Similarity

